Consume Report Generated

Notification Service
Notify user when report completes

Report Generation Request By Orchestrator Service Verify token and Pass for Report Generation

Report Workflow Service
Manage long-running report lifecycle and approval state

Read Report

Report Generation Service
Deterministic report generation → JSON data to Markdown/HTML from existing storage or from AI Generated content

Write Report

Report Generated

Knowledge Sync Service
Coordinate parsing/ chunking/knowledge updates

Document Processing Service
Parsing, normalization, embedding, chunking, mapping Postgres field

Read raw data

Write

Knowledge Store

PostgreSQL

pgvector

GraphRAG

Dump Raw Data

Knowledge Store Service
Manage Vector DB, Graph DB and SQL metadata, embed query and retrieve

Read Only

Consume Data Ingested

Market Data Service
[Scheduled] - Ingest from trusted sources, fetch/ scrape, validate, normalize, store raw/market data, also schedules News + Regulatory ingestion

Data Ingested

Internet sources / API

Google Pub/Sub

LLM Evaluation Service
Offline: Ragas, Deepeval, DeepTeam
Golden Dataset

Agent Service
Orchestrate and execute all ADK agents: Fetch, Financial Expert, Compliance, Industry Researcher, and Consolidation Agents, generate report content, research content

Internet sources

Valkey MemoryStore

MCP and Tool Service
Standardized access to custom and external tools

Research API Service
Receive research request, return from DB - search answer / research report / news

Cache miss Research trigger

Policy and Guardrail Service
Policy enforcement and safety checks
Sign verification token

Guardrails
Input - before
Output - after

Intent Service
Validation, Normalization, Rule-based intent classification + LLM fallback

Research Orchestrator
Coordinate the complete research workflow, report generation trigger, compare companies, Analysis of directors and ownership

Report Generation Request On Approved Content + Verification Token

Metrics

Benchmark Controller Service
Start/manage benchmark runs

Benchmark Verification Service
Collect/analyze metrics and produce verification artifacts

Load Generator
Generate simulated 1M users

Auth Service
JWT + refresh token, checks scope

session/token validation cache

API Gateway
Rate Limiting

HTTPS

Observability Stack

Infra:
For all services
OTel Collector → Prometheus → Grafana

AI/LLM:
For all services making LLM calls
Live: Phoenix (traces) + MLflow (experiments)

Communication

- gRPC
- HTTPS
- Events
- MCP
- Service Mesh (Side car)

Workflows

- Market Data Ingestion
- Knowledge Synchronization
- Research Request Processing
- Multi-Agent Execution
- Report Generation & Approval
- Capacity Validation Benchmark